Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030478

NAME: ADAMA SAMARTH

Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS dd1_db;
USE dd1_db;
```

Output: -

✔	39	15:09:29	CREATE DATABASE IF NOT EXISTS dd1_db	1 row(s) affected	0.110 sec
✔	40	15:09:34	USE dd1_db	0 row(s) affected	0.015 sec

Query 2: -DDL (DATA DEFINITION LANGUAGE)

CREATE TABLE

```
2520030478-Adama
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output: -

✔	41	15:12:39	CREATE TABLE bank_transactions (txn_id INT P...	0 row(s) affected	0.203 sec

Query 3: -ALTER TABLE ADD COLUMN

```
-- Adama Samarth
ALTER TABLE bank_transactions
ADD account_no VARCHAR(20);
```

Output: -

✔	42	15:14:01	ALTER TABLE bank_transactions ADD account_no ...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.093 sec

Query 4: -ALTER TABLE MODIFY COLUMN

```
2520030478-Adama Samarth
ALTER TABLE bank_transactions
MODIFY customer_name VARCHAR(100);
```

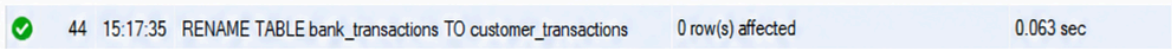
Output: -



Query 5: -RENAME TABLE

```
RENAME TABLE
bank_transactions TO customer_transactions;
```

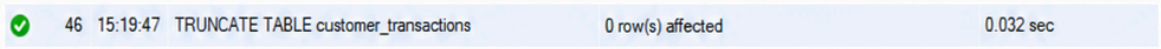
Output: -



Query 6: -TRUNCATE TABLE

```
TRUNCATE TABLE customer_transactions;
```

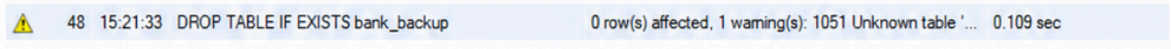
Output: -



Query 7: -DROP TABLE

```
2520030478-Adama Samarth
DROP TABLE IF EXISTS bank_backup;
```

Output: -



Query 8: - DML (DATA MANIPULATION LANGUAGE)

INSERT SAMPLE DATA

```sql
CREATE TABLE customers_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
INSERT INTO customers_transactions VALUES
(101,'Adama Samarth','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Adama Samarth','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Akshara Reddy','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Anirudh Reddy','Vizag','Deposit',8000,'2024-01-10'),
(105,'Yashwanth','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Adithya Vardhan','Vizag','Deposit',15000,'2024-01-12'),
(107,'Ananaya Varshitha','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Varnika Guptha','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Prerka','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Nihita','Vijayawada','Deposit',11000,'2024-01-16');
```

Output: -

✓	52	15:27:20	CREATE TABLE customers_transactions (txn_id I...	0 row(s) affected	0.031 sec
✓	53	15:27:28	INSERT INTO customers_transactions VALUES (101...	10 row(s) affected Records: 10 Duplicates: 0 Warni...	0.015 sec

Query 9: -INSERT

```sql
2520030478-Adama Samarth
INSERT INTO customers_transactions
VALUES
(111,'Venu','Vizag','Deposit',7000,'2024-01-18');
```

Output: -

✓	54	15:28:57	INSERT INTO customers_transactions VALUES (111...	1 row(s) affected	0.062 sec

Query 10: -UPDATE

```sql
UPDATE customers_transactions
SET amount = 5000
WHERE txn_id = 105;
```

Output: -

```
56  15:30:53  UPDATE customers_transactions SET amount = 5000 WHERE tx...  1 row(s) affected  Rows matched: 1  Changed: 1  Warnings: 0  0.015 sec
```

Query 11: -DELETE

```
DELETE FROM customers_transactions
WHERE txn_id = 111;
```

Output: -

```
58  15:32:09  DELETE FROM customers_transactions WHERE txn_id = 111  1 row(s) affected  0.000 sec
```

Query 12: -SELECT

```
SELECT *
FROM customers_transactions;
```

Output: -

```
60  15:33:32  SELECT * FROM customers_transactions LIMIT 0, 1000  10 row(s) returned  0.016 sec / 0.000 sec
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Adama Samarth	Hyderabad	Deposit	5000.00	2024-01-05
102	Adama Samarth	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Akshara Reddy	Vijayawada	Deposit	12000.00	2024-01-08
104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10
105	Yashwanth	Hyderabad	Withdrawal	5000.00	2024-01-11
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
107	Ananaya Varshitha	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
109	Prerka	Vizag	Withdrawal	4000.00	2024-01-15
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Query 13: -SELECT WITH WHERE

```
SELECT *
FROM customers_transactions
WHERE transaction_type='Deposit';
```

Output: -

```
62  15:35:53  SELECT * FROM customers_transactions WHERE transaction_type = 'Deposit'  6 row(s) returned  0.000 sec / 0.000 sec
```

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Adama Samarth	Hyderabad	Deposit	5000.00	2024-01-05
103	Akshara Reddy	Vijayawada	Deposit	12000.00	2024-01-08
104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Adama Samarth	Hyderabad	Deposit	5000.00	2024-01-05
103	Akshara Reddy	Vijayawada	Deposit	12000.00	2024-01-08
104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Query 14: -SELECT WITH ORDER BY

```
SELECT *
FROM customers_transactions
ORDER BY amount DESC;
```

Output: -

64 15:37:59 SELECT * FROM customers_transactions ORDER BY amo... 10 row(s) returned 0.016 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Import:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
103	Akshara Reddy	Vijayawada	Deposit	12000.00	2024-01-08
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14
104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10
101	Adama Samarth	Hyderabad	Deposit	5000.00	2024-01-05
105	Yashwanth	Hyderabad	Withdrawal	5000.00	2024-01-11
109	Prerka	Vizag	Withdrawal	4000.00	2024-01-15
102	Adama Samarth	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Ananaya Varshitha	Vijayawada	Withdrawal	1000.00	2024-01-13
NULL	NULL	NULL	NULL	NULL	NULL

Query 15: - DCL (DATA CONTROL LANGUAGE)

GRANT ALL

```
GRANT ALL PRIVILEGES
ON ddl_db.customers_transactions
TO 'BranchManager'@'localhost';
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

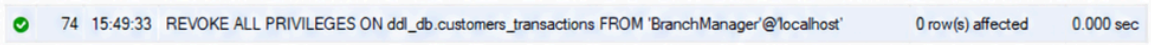
Output: -



Query 16: - REVOKE ALL

```
REVOKE ALL PRIVILEGES
ON ddl_db.customers_transactions
FROM 'BranchManager'@'localhost';
```

Output: -



Query 17: -TCL (TRANSACTION CONTROL LANGUAGE)

COMMIT

```
UPDATE customer_transactions
SET amount=6000
WHERE txn_id=101;

COMMIT;
```

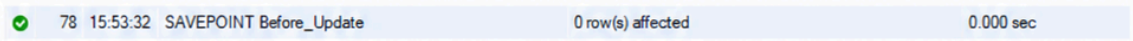
Output: -



Query 18: - SAVEPOINT

```
-- 2520030478-Adama Samarth
SAVEPOINT Before_Update;
```

Output: -



Query 19: - ROLLBACK TO SAVEPOINT

```
-- 2520030478-Adama Samarth
UPDATE customer_transactions
SET amount=99999
WHERE txn_id=102;
ROLLBACK TO Before_Update;
```

Output: -

